

Table 1: Single-point in-place addition comparison with the FLT-in point adder of [JSB+25]

n	Strategy	#Toffoli	#CNOT	Width	Toffoli depth	NCT depth	DW-cost	TDW-cost
163	[JSB+25]	193354	2540458	53134	46	5685	1.13×2^{28}	1.17×2^{21}
	Our Implementation	193354	2417882	53134	46	5888	1.17×2^{28}	1.17×2^{21}
233	[JSB+25]	303970	4516616	82899	50	7059	1.09×2^{29}	1.98×2^{21}
	Our Implementation	303970	4326952	82899	50	7186	1.11×2^{29}	1.98×2^{21}
283	[JSB+25]	534762	7856986	144672	54	10957	1.48×2^{30}	1.86×2^{22}
	Our Implementation	534762	7563802	144672	54	11149	1.50×2^{30}	1.86×2^{22}
571	[JSB+25]	1871402	30657812	500450	62	23596	1.37×2^{33}	1.85×2^{24}
	Our Implementation	1871402	29759716	500450	62	23766	1.38×2^{33}	1.85×2^{24}